

Section 6.1 Worksheet

1. For $T(x, y) = (2x - y, x + 3y)$, find $T(4, -2)$.
2. Find the preimage of $(5, 6)$ under the transformation in Problem 1.
3. Verify that $T(x, y) = (x - 2y, 3x + y)$ is linear.
4. Explain why $F(x, y) = (x + 1, y)$ is not linear.
5. Suppose $T(\mathbf{e}_1) = (9, 1)$, $T(\mathbf{e}_2) = (20, -3)$, and $T(\mathbf{e}_3) = (21, 4)$. Find $T(1, -2, 3)$.
6. Find $D(3 - 2x + 5x^2 - x^3)$ and evaluate $\int_0^1 (1 + x) dx$.

Section 6.2 Worksheet

1. Find the kernel and range of $T(x, y, z) = (x, y, 0)$.
2. For $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$, find a basis for $\ker(A)$ and the rank of A .
3. Verify rank-nullity for the matrix in Problem 2.
4. Determine whether the transformation in Problem 2 is one-to-one, onto, both, or neither.
5. Show that $C(a + bx + cx^2) = (a, b, c)$ is an isomorphism from P_2 to $\mathbb{R}^3$.
6. Explain why the kernel of every linear transformation contains the zero vector.

Section 6.3 Worksheet

1. Find the standard matrix for $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$.
2. Use the matrix from Problem 1 to find $T(1, 2, -1)$.
3. Let $A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find A_2A_1 and A_1A_2 .
4. Find the inverse transformation for $T(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \mathbf{x}$.
5. Find the matrix of $T(x, y) = (x + y, 2x - y)$ relative to $B = \{(1, 1), (1, -1)\}$ and $C = \{(1, 0), (1, 1)\}$.
6. Write the matrix of differentiation $D : P_3 \rightarrow P_2$ relative to the standard polynomial bases.

Section 6.4 Worksheet

1. For $B = \{(1, 1), (1, -1)\}$ and $C = \{(1, 0), (1, 1)\}$, find $P_{C \leftarrow B}$.
2. Convert $[\mathbf{x}]_B = (3, 2)$ to C -coordinates.
3. For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, compute $P^{-1}AP$.
4. Verify that A and $P^{-1}AP$ have the same trace and determinant.
5. Use the diagonal representation to compute A^5 .
6. Explain why the order of vectors in a basis matters.

Section 6.5 Worksheet

1. Identify the transformations represented by $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.
2. Reflect $(3, -2)$ across the line $y = x$.
3. Apply $H(1/2)$ to the triangle with vertices $(0, 0)$, $(4, 0)$, and $(0, 2)$.
4. Rotate $(2, 0)$ counterclockwise by 30° .
5. Find the matrix for a horizontal stretch by 2 followed by a 90° counterclockwise rotation.
6. Rotate $(1, 2, 3)$ counterclockwise by 90° about the z -axis.